

Rover Evidence Lab

GROW • Science • 40 minutes

I can carry out a comparative fair test, use repeated distance results for a conclusion and improve the method.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
repeat	same method again
conclusion	what the results support
limitation	what the test cannot show

First idea

Smooth-surface results are 48, 51 and 49 cm. What should happen to the 51 cm result?

- A. Keep it with the other valid repeats
- B. Delete it because it is not identical
- C. Keep only 51 because it is the furthest

My first idea and reason:

“I predict the rover goes farther on smooth, because...”

a prediction is held, not marked

Prediction

We do • choose, explain, check

Which recording choice gives the strongest evidence?

- A. Record all three valid results for each condition using the same measuring rule.
- B. Keep only the furthest result from each condition.
- C. Change a result when it does not match the prediction.

My choice: _____ Evidence or reason:

Rover Results Lab

Place each evidence card under Smooth, Fabric or Conclusion. Freeze the pattern before choosing an improvement.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
SMOOTH	
FABRIC	
CONCLUSION / LIMIT	

Activity cards

1 • 48, 51, 49 cm	2 • Range: 3 cm (48–51 cm interval)
Three digital-twin runs on the smooth surface.	Maximum minus minimum for the smooth results: $51 - 48 = 3$ cm.
3 • 27, 29, 28 cm	4 • Range: 2 cm (27–29 cm interval)
Three digital-twin runs on the fabric surface.	Maximum minus minimum for the fabric results: $29 - 27 = 2$ cm.
5 • Smooth produced greater travel distance	6 • One classroom rover is a simplified model
Lower friction opposed the rover's motion less in this model.	Real rovers face different gravity, terrain and designs.

State the pattern, one evidence-linked conclusion, one method improvement and one model limitation.

Your lesson record

Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation.

Model helps us see

The fixed six-result set provides a complete parity route and models how raw data become a cautious conclusion.

Model does not show

A short classroom ramp tests one rover, two surfaces and one set of conditions; it cannot represent gravity, terrain or every real rover design.

Supported

Direct each run, match the six values to conditions and complete an oral or symbol-supported conclusion.

Adult handles equipment; use pre-labelled rows and the frame: On ____ it went ____ because ____.

Standard

Complete three safe repeats per condition, record all results and write a conclusion plus one improvement.

Check the method against change, measure, keep and repeat before concluding.

Stretch

Compare typical values, calculate numerical ranges, explain variation and evaluate transfer to a real rover.

Suggest one justified method improvement without changing the original raw data.

Evidence target

Prediction, all six raw results, pattern statement, evidence-linked conclusion, one improvement and one model limitation.

Exit, Audience and evidence • W13B

Supported: Which condition gave the greater travel distance?

Standard: Use one repeated result pattern to explain the effect of surface on the rover.

Stretch: Give one method improvement and one model limitation, explaining why each matters.

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.

SMOOTH	FABRIC
48 cm · 51 cm · 49 cm mean $\approx$ 49 cm	27 cm · 30 cm · 28 cm mean $\approx$ 28 cm

Results

Pattern: every fabric run is shorter.

Conclusion: more friction on fabric → shorter distance.

Improvement: a release gate so every start is identical.

Conclusion

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
